

FUA'AMOTU, Tonga

WMO: 917920

Lat: 21.241S

Lon: 175.150W

Elev: 125

StdP: 14.63

Time Zone: 12.00 (PCF)

Period: 94-19

WBAN: 00487

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	59.4	62.4	51.8	57.4	67.4	53.6	61.3	67.7	21.1	72.8	19.6	72.9	0.8	150	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.9	87.7	79.8	86.3	79.0	85.8	78.6	80.9	85.2	80.1	84.5	79.4	83.6	8.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	152.5	83.5	79.0	151.0	83.2	78.5	148.5	82.6	44.7	86.2	43.9	85.1	43.1	83.7	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	17.7	15.9	55.5	89.4	2.4	1.5	53.8	90.5	52.3	91.4	51.0	92.2	49.2	93.3		
(5)			54.1	82.0	2.1	1.3	52.5	82.9	51.3	83.7	50.1	84.4	48.6	85.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.6	79.6	80.4	79.8	77.9	75.1	73.0	71.2	71.1	71.9	73.7	76.0	78.2
	DBStd	4.18	2.10	1.98	2.00	2.28	2.56	2.82	2.75	2.79	3.18	2.98	2.57	2.36
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	0	0	0	0	0	0	0	1	1	0	0	0
	CDD50	9354	917	850	923	838	778	689	658	655	657	736	780	873
	CDD65	3882	452	430	458	388	313	239	194	191	208	271	330	408
	CDH74	24408	3967	4028	4026	2758	1415	713	403	398	583	1077	1905	3135
	CDH80	4408	903	1038	918	442	114	33	8	10	28	89	229	596
(14) Wind	WSAvg	7.7	7.8	7.6	7.1	7.7	7.5	7.6	7.7	7.8	7.8	7.8	7.7	8.0
(15) Precipitation	PrecAvg	68.2	7.9	8.7	7.3	6.3	4.0	4.0	4.3	4.9	4.6	4.3	3.6	5.7
	PrecMax	102.0	20.9	26.1	16.3	13.0	12.5	9.4	9.3	14.4	14.6	19.7	10.4	12.3
	PrecMin	36.5	0.2	1.7	1.1	0.7	1.0	0.7	1.0	0.6	0.8	0.9	0.8	0.4
	PrecStd	20.1	6.4	6.6	3.7	4.0	3.3	2.5	2.4	3.4	3.5	4.3	2.7	2.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	89.5	88.1	87.2	84.4	82.6	80.9	81.0	82.6	84.2	85.4	87.1
		MCWB	79.7	80.0	80.2	78.6	77.2	76.3	73.9	74.7	75.6	76.5	77.5	79.0
	2%	DB	86.4	87.8	87.0	85.7	82.6	80.7	79.0	79.1	80.5	82.3	84.0	85.8
		MCWB	79.0	79.9	79.6	78.5	76.4	74.7	72.9	73.0	74.1	75.5	76.6	78.0
	5%	DB	85.9	86.3	85.9	84.1	81.0	79.1	77.4	77.4	78.8	80.7	82.5	84.4
		MCWB	78.6	79.2	79.0	77.3	74.6	73.4	71.5	71.3	72.6	74.0	75.3	76.9
	10%	DB	84.4	85.6	84.5	82.6	79.8	77.4	76.3	76.3	77.1	79.0	81.0	83.4
		MCWB	77.6	78.8	77.9	76.0	73.6	72.0	70.5	70.4	71.0	72.5	74.1	76.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.8	81.9	81.8	80.2	79.2	78.0	76.3	76.4	77.1	78.1	79.2	80.3
		MCDB	85.7	85.9	85.9	84.3	82.5	81.0	78.9	79.6	80.8	82.3	83.2	84.8
	2%	WB	80.4	80.9	80.7	79.2	78.0	76.4	74.9	74.8	75.4	76.6	78.1	79.1
		MCDB	84.7	85.5	85.1	83.4	81.2	79.4	77.4	77.8	78.8	80.7	81.9	83.8
	5%	WB	79.5	80.0	79.8	78.4	76.6	75.2	73.6	73.5	74.2	75.4	77.1	78.3
		MCDB	83.7	84.6	84.0	82.5	79.6	78.0	76.3	76.5	77.5	79.0	80.8	82.8
	10%	WB	78.8	79.2	79.0	77.6	75.5	73.8	72.2	72.0	72.8	74.2	76.1	77.6
		MCDB	82.9	83.5	83.0	81.3	78.5	76.7	75.1	75.1	76.1	77.8	79.7	81.8
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	9.9	9.7	9.4	9.4	9.3	9.8	9.7	10.1	10.2	10.1	10.1
		MCDBR	11.2	11.2	11.1	11.0	10.6	11.0	11.4	11.2	11.5	11.4	11.2	11.0
	5% WB	MCWBR	5.8	5.7	5.9	6.0	6.4	6.6	7.1	7.0	7.0	6.5	6.0	5.9
		MCWBR	10.0	10.4	10.0	10.2	9.6	10.0	9.5	10.4	10.1	10.3	9.8	10.1
(40) Clear-Sky Solar Irradiance	taub		0.384	0.376	0.378	0.390	0.371	0.369	0.358	0.361	0.381	0.369	0.371	0.379
		taud	2.611	2.645	2.645	2.588	2.619	2.607	2.626	2.618	2.551	2.583	2.601	2.597
	Ebn at Noon		305	303	295	278	272	266	273	283	289	302	306	306
		Edh at Noon	33	31	30	30	27	26	26	28	32	33	33	33
(44) All-Sky Solar Radiation	RadAvg		2025	1951	1729	1455	1232	1086	1172	1357	1619	1891	2046	2128
	RadStd		186	171	119	163	78	72	71	120	126	130	160	119

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+1.87	N/A	+1.28	+1.40	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	+1.87	N/A	+1.28	+1.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page